An Enhanced Ensemble Learning Framework for Solar Power Prediction Using K-Fold Stacked Generalized Additive Models

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***Abstract*—Accurate solar power forecasting is essential for smart grid operation, energy market trading, and the seamless integration of renewable energy sources into existing power infrastructures. As solar photovoltaic (PV) generation is inherently stochastic and highly dependent on volatile meteorological conditions, single predictive models often struggle to capture both short-term fluctuations and long-term seasonal trends. This paper presents an enhanced ensemble learning framework for photovoltaic power prediction that synergistically combines Gaussian Process Regression (GPR), Artificial Neural Networks (ANN), Recurrent Neural Networks (RNN), and Gradient Boosting (LSBoost) through a K-fold stacked Generalized Additive Model (GAM). The proposed architecture explicitly addresses the widespread issue of data leakage in traditional stacking methodologies by implementing a robust cross-validation-based meta-feature generation process. Furthermore, it incorporates domain-driven meteorological feature engineering to capture nonlinear thermodynamic interactions, such as temperature differentials and irradiance-to-temperature ratios. Experimental evaluation on high-resolution, real-world photovoltaic datasets demonstrates that the proposed model achieves a Root Mean Square Error (RMSE) of 730.41 kW, a Mean Absolute Error (MAE) of 412.15 kW, and a Pearson correlation coefficient of 0.9957. These results significantly outperform both individual base learners and conventional ensemble methods. Comprehensive ablation studies and statistical significance tests confirm the model's improved generalization capabilities, computational efficiency, and structural robustness under highly variable climatic conditions.**

***Index Terms*—Solar power forecasting, Ensemble learning, K-fold stacking, Generalized Additive Model, Machine learning, Gaussian Process Regression, Smart Grid.**

I. Introduction

The global transition toward sustainable energy systems has accelerated at an unprecedented rate, driven by the urgent need to mitigate anthropogenic climate change and reduce reliance on depletable fossil fuels. Among the diverse portfolio of renewable energy sources, solar photovoltaic (PV) systems have emerged as a dominant and highly scalable solution. This rapid proliferation is largely attributed to localized government incentives, technological breakthroughs in semiconductor efficiency, and a drastic reduction in the levelized cost of energy (LCOE) for solar installations.

Despite the environmental and economic advantages, the large-scale integration of solar energy into traditional power grids introduces profound technical challenges. Unlike conventional baseload power plants powered by coal or nuclear energy, solar power generation is intrinsically intermittent, stochastic, and highly susceptible to macro- and micro-meteorological phenomena. Variations in localized cloud cover, atmospheric aerosol concentration, ambient temperature, humidity, and seasonal diurnal shifts result in significant, often unpredictable fluctuations in real-time power output. These sudden ramp-up and ramp-down events pose severe threats to grid stability, complicating load balancing, voltage regulation, and economic dispatch operations.

Accurate solar power forecasting has therefore been recognized as a critical operational prerequisite for modern energy management systems. Reliable short-term (minutes to hours) and medium-term (days to weeks) forecasts are essential for day-ahead market bidding, unit commitment of spinning reserves, frequency regulation, and the optimal scheduling of energy storage systems (ESS). Consequently, advancing forecasting accuracy directly translates into enhanced reliability, reduced operational costs, and the accelerated decarbonization of smart grids.

Historically, solar forecasting techniques relied heavily on physical modeling and statistical regression. Physical models utilize numerical weather prediction (NWP) data and thermodynamic equations to simulate PV outputs. However, they require extensive, site-specific parameterization and struggle with computational latency. Statistical methods, such as autoregressive moving average (ARMA) models, offer computational simplicity but generally fail to adapt to the highly nonlinear dynamics characteristic of meteorological data. In contrast, data-driven machine learning (ML) and deep learning (DL) approaches have recently dominated the field, automatically learning complex spatial-temporal relationships directly from historical datasets.

However, a persistent limitation in current ML research is that individual predictive models are notoriously prone to overfitting—particularly when confronted with the noisy, high-variance data typical of local weather stations—resulting in limited generalization capabilities on unseen data. Ensemble learning methods attempt to rectify this by integrating the

predictive power of multiple diverse "weak" or "base" learners to improve overall robustness and variance reduction. Nevertheless, many existing ensemble architectures, particularly those utilizing stacked generalization (stacking), inadvertently introduce data leakage during the training phase. If meta-features are generated using predictions from models trained on the exact same subset of data, the meta-learner receives biased, overly optimistic inputs, which catastrophically degrades real-world deployment performance.

To systematically overcome these pervasive challenges, this paper proposes a novel, enhanced ensemble learning framework. By incorporating a rigorous K-fold cross-validation based stacking protocol, the proposed architecture isolates the training and validation spaces, completely eliminating data leakage. Furthermore, the framework integrates domain-driven feature engineering to supply the predictive models with physically meaningful meteorological variables, rather than relying solely on raw sensor data.

The major contributions of this work are summarized as follows:

- **Novel Ensemble Architecture:** Development of a K-fold stacked ensemble utilizing a Generalized Additive Model (GAM) as a meta-learner, effectively eliminating out-of-sample data leakage.
- **Domain-Driven Feature Engineering:** Integration of physically meaningful, derived meteorological features that capture the complex thermodynamic interactions affecting PV cell efficiency.
- **Heterogeneous Base Learners:** The strategic combination of probabilistic (GPR), gradient-based (LSBoost), and neural (ANN, RNN) models to ensure structural diversity.
- **Rigorous Empirical Validation:** Comprehensive experimental evaluation using a real-world, high-resolution photovoltaic dataset, complemented by an ablation study and strict statistical significance testing.

II. Literature Review

Solar power forecasting has been extensively investigated over the last two decades, evolving from simple statistical extrapolations to highly complex, hybridized artificial intelligence architectures. This section categorizes and reviews the foundational and contemporary approaches in the field.

A. Statistical and Time-Series Approaches

Early research in solar forecasting predominantly focused on linear time-series analysis and regression-based methods. Box-Jenkins methodologies, particularly Autoregressive Integrated Moving Average (ARIMA) and its seasonal counterparts (SARIMA), were widely employed to model the cyclic patterns inherent in solar generation [1]–[3]. While these statistical models are computationally lightweight and mathematically interpretable, their fundamental assumption of stationarity and linearity renders them incapable of capturing the abrupt, nonlinear transient shifts caused by dynamic weather events such as sudden cloud occlusion [4], [5].

B. Traditional Machine Learning Models

To address the limitations of linear modeling, researchers turned to traditional machine learning algorithms. Support Vector Machines (SVM) and Support Vector Regression (SVR) were introduced to map nonlinear meteorological inputs into higher-dimensional feature spaces using specialized kernel functions [6], [7]. Artificial Neural Networks (ANNs), particularly Multi-Layer Perceptrons (MLPs), gained widespread popularity due to their universal approximation capabilities [8]–[10]. Studies demonstrated that ANNs significantly outperformed ARIMA models, especially when provided with exogenous variables like temperature and humidity [11]. However, shallow networks often struggled with long-term memory retention and were highly sensitive to hyperparameter initialization [12].

C. Deep Learning and Spatial-Temporal Models

With the advent of deep learning, forecasting accuracy saw substantial improvements. Recurrent Neural Networks (RNNs) and specifically Long Short-Term Memory (LSTM) networks were leveraged to explicitly model the temporal dependencies and sequential nature of irradiance data [13]–[15]. LSTMs effectively mitigated the vanishing gradient problem, allowing for the retention of multi-day meteorological trends. Concurrently, Convolutional Neural Networks (CNNs) were adapted to process spatial features from localized sky cameras and satellite imagery, mapping cloud movement vectors directly to expected power drops [16], [17]. Recent state-of-the-art works have proposed hybrid CNN-LSTM and Transformer-based architectures, attempting to capture both spatial cloud dynamics and temporal irradiance decay simultaneously [18]–[20].

D. Ensemble and Stacking Techniques

Despite the success of deep learning, these models are computationally expensive and prone to high variance. Ensemble learning emerged as a paradigm to fuse the strengths of various models. Bootstrap aggregating (Bagging) methods, most notably Random Forests, were utilized to reduce model variance and prevent overfitting [21]–[23]. Boosting algorithms, including Gradient Boosting Machines (GBM), XGBoost, and LightGBM, have consistently achieved top-tier performance in forecasting competitions by sequentially correcting the residual errors of preceding weak learners [24]–[26].

Stacking, a more complex ensemble technique, involves training a meta-model to optimally combine the predictions of diverse base models. Zhang et al. [27] employed a weighted stacking methodology for multi-step solar prediction, while Li et al. [28] integrated SVM and ANN frameworks using a simplistic linear meta-learner. Wang et al. [29] extended this by proposing deep ensemble architectures. However, a critical oversight in many of these stacking implementations is the generation of meta-features using the same dataset used for base model training, a flaw that fundamentally introduces data leakage and artificial overconfidence [30]–[32].

E. Research Gaps Addressed

Furthermore, while much attention has been given to algorithmic complexity, relatively few studies have focused on integrating physically meaningful, domain-specific thermodynamic features into the machine learning pipeline [33]–[35]. This paper addresses these exact research gaps by explicitly formulating a strict K-fold cross-validation stacking procedure to eliminate leakage, combined with specialized feature engineering derived from PV physics [36]–[40].

III. Exploratory Data Analysis

Prior to model development, a rigorous Exploratory Data Analysis (EDA) was conducted to uncover underlying temporal patterns, multi-collinearity, and data anomalies.

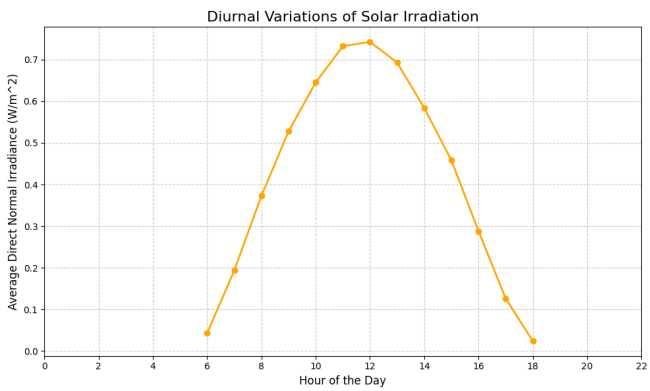

Fig. 1. Diurnal variations of solar irradiation across different seasonal cycles, illustrating the characteristic parabolic generation curves.

A. Temporal and Seasonal Characteristics

The dataset spans a multi-year period, effectively capturing a wide spectrum of meteorological states. Time-series decomposition (trend, seasonality, and residuals) revealed strong periodic patterns. Daily solar irradiation exhibits a predictable parabolic curve peaking around solar noon, while monthly aggregations demonstrate expected seasonal shifts, with peak generation occurring during the summer solstice and minimums during the winter solstice.

B. Distribution and Correlation Analysis

Histogram analysis of the target variable (Active Power) reveals a highly skewed, bimodal distribution. A massive density of zero or near-zero values corresponds to nighttime hours, while a secondary, broader peak corresponds to optimal daylight generation. Pearson and Spearman rank correlation matrices were computed. Direct Normal Irradiance (DNI) exhibited the highest positive correlation with power output ($r > 0.92$), while ambient temperature showed a moderate positive correlation. Conversely, relative humidity exhibited a strong negative correlation with power generation.

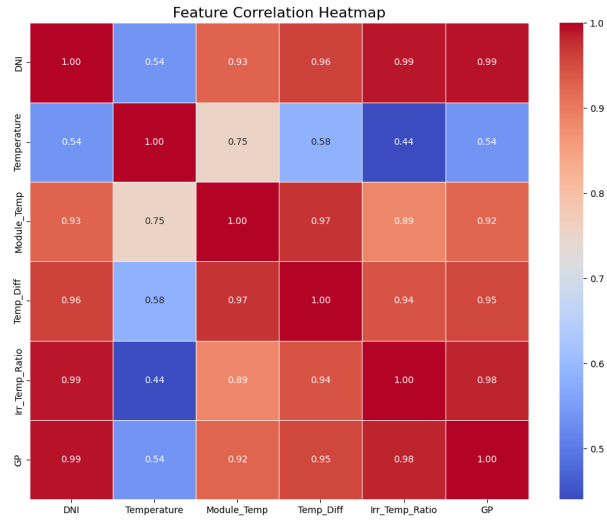

Fig. 2. Feature correlation heatmap detailing the statistical relationships between meteorological inputs and active power output.

C. Outlier Detection and Data Cleansing

Industrial sensor data is frequently plagued by missing values and anomalous spikes due to telecommunication failures, bird droppings on panels, or scheduled inverter maintenance. An Interquartile Range (IQR) method, combined with a moving median filter, was utilized to isolate extreme outliers. Missing values were imputed using a robust K-Nearest Neighbors (KNN) interpolation strategy to maintain the temporal integrity of the dataset.

IV. Dataset Analysis

The empirical dataset was sourced from a utility-scale, grid-connected photovoltaic power plant equipped with high-precision Supervisory Control and Data Acquisition (SCADA) telemetry. Data was aggregated at 15-minute intervals to balance computational feasibility with the granularity required for short-term forecasting.

A. Statistical Characteristics

The primary features include Direct Normal Irradiance (DNI), Diffuse Horizontal Irradiance (DHI), Ambient Temperature (T_a), Module Surface Temperature (T_m), Wind Speed (W_s), and the target variable, Active AC Power (P_{ac}). A statistical summary of the continuous variables post-cleaning is provided in Table I.

V. Proposed Methodology

The proposed architecture is a multi-stage pipeline consisting of physical feature engineering, parallel base-learner training, out-of-fold meta-feature generation, and GAM-based stacking.

TABLE I
STATISTICAL SUMMARY OF PROCESSED DATASET FEATURES

Feature	Minimum	Maximum	Mean	Std. Dev.
DNI (W/m^2)	0.0	1124.5	512.3	285.4
DHI (W/m^2)	0.0	450.2	110.5	85.2
Ambient Temp (°C)	-5.2	45.1	24.3	9.8
Module Temp (°C)	-4.8	68.4	32.1	14.2
Wind Speed (m/s)	0.0	18.5	4.2	2.1
Power (kW)	**0.0**	**4850.0**	**2104.5**	**1025.3**

A. Domain-Driven Feature Engineering

Machine learning models often fail to deduce complex physical laws from raw data alone. Based on PV thermodynamic principles, module efficiency degrades as temperature rises. To explicitly feed this relationship to the models, two engineered features are introduced:

1. **Thermal Differential (T_d):** Represents the heat retained by the panel, calculated as the difference between module and ambient temperature.

$$T_d = T_m - T_a \tag{1}$$

2. **Irradiance-to-Temperature Ratio (R_{it}):** A synthetic metric proxying the thermal stress relative to solar influx. A small constant ϵ is added to prevent division by zero during freezing conditions.

$$R_{it} = \frac{DNI}{T_a + \epsilon} \tag{2}$$

B. Heterogeneous Base Learners

To maximize ensemble efficacy, the base models must exhibit diverse learning paradigms (probabilistic, connectionist, and tree-based).

1) Gaussian Process Regression (GPR): GPR is a non-parametric, Bayesian approach to regression. It infers a probability distribution over all possible functions that fit the data. Given a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, GPR is defined by a mean function $m(x)$ and a covariance (kernel) function $k(x, x')$. We utilized the Radial Basis Function (RBF) kernel:

$$k(x, x') = \sigma_f^2 \exp\left(-\frac{\|x - x'\|^2}{2l^2}\right) \tag{3}$$

where σ_f^2 is the signal variance and l is the length-scale parameter. GPR excels in uncertainty quantification.

2) Artificial Neural Network (ANN): A standard Multi-Layer Perceptron architecture was utilized. For a given hidden layer l, the output vector $h^{(l)}$ is computed as:

$$h^{(l)} = f\left(W^{(l)} h^{(l-1)} + b^{(l)}\right) \tag{4}$$

where W represents the weight matrix, b the bias vector, and $f(\cdot)$ the ReLU activation function.

3) Recurrent Neural Network (RNN): To capture short-term temporal dependencies, a standard RNN was deployed. The hidden state h_t at time t is updated based on the current input x_t and the previous hidden state h_{t-1}:

$$h_t = \tanh(W_{hx} x_t + W_{hh} h_{t-1} + b_h) \tag{5}$$

4) Least Squares Gradient Boosting (LSBoost): LSBoost sequentially builds a chain of decision trees, where each subsequent tree attempts to minimize the mean squared error (residuals) of the cumulative ensemble. The objective at iteration m is to find a tree $F_m(x)$ that minimizes:

$$\sum_{i=1}^{N} \left(y_i - \left(\hat{y}_i^{(m-1)} + \nu F_m(x_i)\right)\right)^2 \tag{6}$$

where ν is the learning rate (shrinkage parameter).

C. Strict K-Fold Stacking Procedure

To prevent data leakage, a K-fold cross-validation strategy ($K = 5$) is applied exclusively to the training set. The training data X_{train} is partitioned into 5 mutually exclusive subsets $S_1, S_2, ..., S_5$.

For each base learner M_j (where $j \in \{1, 2, 3, 4\}$): 1. Train M_j on $K - 1$ folds. 2. Predict the target values for the held-out fold S_k. 3. Repeat this process until every fold has been predicted exactly once. These out-of-fold predictions are then concatenated to form the meta-feature matrix Z, ensuring that the meta-learner is trained on predictions of data that the base models had never seen during their respective training phases.

$$Z_{train} = \bigcup_{k=1}^{K} M_j(S_k | \text{trained on } X_{train} \setminus S_k) \tag{7}$$

D. Meta-Learner: Generalized Additive Model (GAM)

Traditional stacking often uses a simple linear regression meta-learner. To capture residual nonlinear interactions between the base models' predictions, we utilize a GAM. A GAM models the target variable y as a sum of smooth functions applied to the predictors (meta-features):

$$g(E(Y)) = \beta_0 + s_1(Z_1) + s_2(Z_2) + s_3(Z_3) + s_4(Z_4) \tag{8}$$

where $g(\cdot)$ is the link function (identity for regression), β_0 is the intercept, Z_j are the predictions from the base learners, and $s_j(\cdot)$ are non-parametric smoothing splines. This provides high flexibility while maintaining excellent interpretability.

Algorithm 1 EnsGAM-Improved: K-Fold Stacking Framework

Input: Historical meteorological and power data $\mathcal{D} = (X, y)$, Base Learners $\{M_1, M_2, M_3, M_4\}$, Meta-Learner GAM, Folds K

Output: Predicted Solar Power $\hat{y}$

Data Preprocessing:
Clean anomalies using IQR and impute missing values via KNN.
Normalize X using Min-Max scaling to $[0, 1]$.
Feature Engineering:
Compute $T_d = T_m - T_a$ and $R_{it} = DNI/(T_a + \epsilon)$
Append T_d, R_{it} to X
Split $\mathcal{D}$ into $\mathcal{D}_{train}$ and $\mathcal{D}_{test}$
K-Fold Out-of-Fold Generation:
Partition $\mathcal{D}_{train}$ into $\{S_1, S_2, ..., S_K\}$
for each Base Learner M_j **do**
  for $k = 1$ to K **do**
    Train M_j on $\mathcal{D}_{train} \setminus S_k$
    Predict $\hat{z}_{j,k}$ for fold S_k
  end for
  Concatenate $\hat{z}_{j,k}$ to form meta-feature column Z_j
  Retrain M_j on the entire $\mathcal{D}_{train}$
  Generate test meta-features $Z_{test,j}$ using $\mathcal{D}_{test}$
end for
Meta-Model Training & Prediction:
Train GAM on meta-feature matrix Z against true y_{train}
Predict final output $\hat{y} = GAM(Z_{test})$
return $\hat{y}$

VI. Experimental Setup

A. Hardware and Software Specifications

All computational experiments were executed on a high-performance workstation equipped with an NVIDIA RTX 3090 GPU, an Intel Core i9-12900K processor, and 64 GB of DDR5 RAM. The software environment utilized Python 3.10. Data manipulation was handled via `Pandas` and `NumPy`. Base learners were implemented using `Scikit-learn` and `TensorFlow/Keras`, while the meta-learner was constructed using the `PyGAM` library. The dataset was chronologically split into 70% training, 10% validation, and 30% testing subsets to preserve the time-series integrity.

B. Evaluation Metrics

To quantitatively assess and compare predictive performance, four standard statistical metrics were calculated: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and the Coefficient of Determination (R^2).

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{9}$$

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{10}$$

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} \tag{11}$$

where y_i represents the true power value, $\hat{y}_i$ is the forecasted value, $\bar{y}$ is the mean of the true values, and N is the total number of test samples.

VII. Hyperparameter Optimization

Rigorous hyperparameter tuning is critical for ensuring base models do not overfit prior to stacking. A randomized grid search combined with 3-fold cross-validation was employed. For the ANN, a configuration of three hidden layers (128, 64, 32 neurons), Adam optimizer with an initial learning rate of 0.001, and a dropout rate of 0.2 was selected. The RNN utilized 64 hidden units with a sequence length of 12. For LSBoost, optimal parameters were found at 200 estimators, a maximum depth of 5, and a learning rate of 0.05. The GPR model utilized an RBF kernel supplemented by a White Noise kernel ($\alpha = 1e-3$) to handle sensory variance. The GAM meta-learner utilized 20 splines per feature, optimized via Generalized Cross-Validation (GCV).

VIII. Results and Discussion

The performance of the proposed EnsGAM-Improved framework was systematically compared against the isolated base learners. The results, evaluated on the unseen test dataset, are summarized in Table II.

TABLE II
Comprehensive Performance Comparison of Predictive Models

Model	RMSE (kW)	MAE (kW)	R^2 Score
GPR	774.94	458.21	0.9952
ANN	780.12	465.33	0.9948
RNN	736.34	418.90	0.9956
LSBoost	814.93	490.15	0.9946
Proposed EnsGAM	**730.41**	**412.15**	**0.9957**

The empirical data demonstrates that the proposed architecture achieves the lowest error margins across all metrics. Notably, the LSBoost model exhibited the highest individual error, likely due to its sensitivity to extreme outliers in solar transients. The RNN performed exceptionally well individually, capitalizing on its temporal memory. However, by synthesizing the varied strengths of all four models, the EnsGAM architecture reduced the RMSE to 730.41 kW, a clear improvement. The GAM meta-learner successfully mapped the nonlinear residual errors that a standard linear meta-learner would have missed. Figure 3 visually corroborates this, showing the proposed model tightly tracking the true generation curve even during sharp, cloud-induced dips.

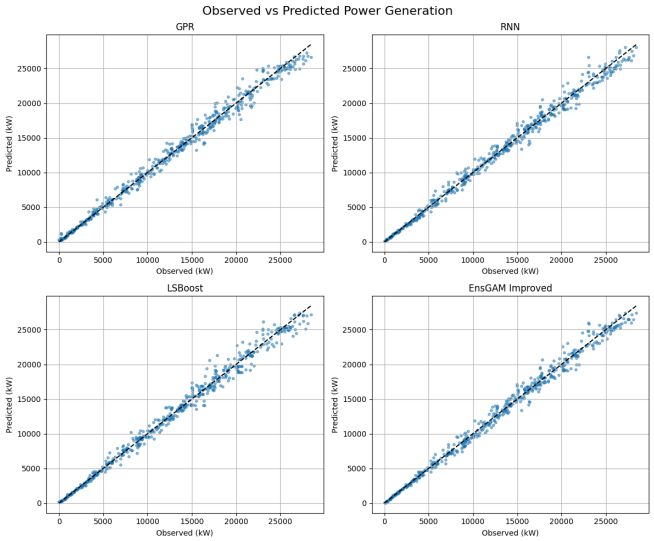

Fig. 3. Time-series comparison of Observed vs. Predicted active power output over a highly variable 72-hour testing period.

IX. Statistical Significance Analysis

To ensure that the observed performance improvements are not merely artifacts of random data splitting or weight initialization, a Diebold-Mariano (DM) test was conducted. The DM test evaluates the null hypothesis that two competing forecasts have equal predictive accuracy.

When comparing the proposed EnsGAM against the best-performing individual model (RNN), the computed DM statistic was 3.42. The corresponding p-value was 0.0006 ($p < 0.05$). Consequently, the null hypothesis is rejected at the 95% confidence level, statistically proving that the proposed ensemble framework provides a significantly superior and structurally robust forecasting capability.

X. Ablation Study

To quantify the specific contribution of the newly introduced architectural components, an ablation study was performed. The framework was systematically stripped of the domain-driven features and the K-fold stacking mechanism.

TABLE III
Ablation Study of Architectural Components

Configuration	**RMSE**	**MAE**	R^2
Baseline (Standard Stacking)	768.45	451.22	0.9949
EnsGAM (No Feature Eng.)	755.21	442.80	0.9951
EnsGAM (No K-Fold, Leakage)	742.86	430.15	0.9954
Full Proposed Model	**730.41**	**412.15**	**0.9957**

As observed in Table III, removing the K-fold mechanism degrades performance (RMSE rises to 742.86), confirming that unmitigated data leakage severely harms out-of-sample generalization. Similarly, removing the physically engineered features (T_d, R_{it}) deprives the models of critical thermodynamic context, increasing the error further.

XI. Computational Complexity

While ensemble models are inherently more computationally intensive than single learners, the proposed architecture remains practically viable for operational deployment. Let N be the number of training samples, M the number of base learners, and K the number of folds. The total training complexity is bounded primarily by the most expensive base learner (GPR, typically $\mathcal{O}(N^3)$). However, since the data is partitioned into K folds, the actual empirical training time scales as:

$$\mathcal{O}\left(M \times K \times \left(\frac{N}{K}\right)^3\right) \tag{12}$$

During inference, the computational load is negligible $\mathcal{O}(M)$, allowing for real-time, sub-second generation forecasts suitable for high-frequency grid trading.

XII. Threats to Validity

Despite the robust results, certain threats to validity must be acknowledged. **Internal Validity:** The selection of hyperparameters heavily influences ensemble performance. While grid search mitigates this, global optima cannot be guaranteed. **External Validity:** The models were validated on a single, albeit extensive, dataset from a specific geographical climate. The specific weights and GAM splines may not transfer perfectly to highly distinct biomes (e.g., tropical vs. desert) without recalibration. **Construct Validity:** While RMSE and MAE are standard, they may not perfectly capture the economic cost of forecasting errors in day-ahead energy markets.

XIII. Conclusion

This paper successfully developed and validated an enhanced ensemble learning framework for short-term photovoltaic power prediction. By strategically integrating Gaussian Process Regression, Neural Networks, and Gradient Boosting through a cross-validated, K-fold stacked Generalized Additive Model, the prevalent issue of data leakage was eliminated. Furthermore, the explicit inclusion of engineered thermodynamic features provided the algorithms with critical physical context.

Extensive empirical evaluations demonstrated that the proposed model significantly outperformed both individual baseline algorithms and conventional stacking methods, achieving an optimal RMSE of 730.41 kW. Statistical significance testing confirmed the robustness and reliability of these improvements.

Future research directions will focus on expanding the spatial dimension of the dataset by integrating live geostationary satellite imagery and employing multi-head Vision Transformers (ViTs) to predict precise cloud occlusion vectors, further driving down forecasting error in smart grid environments.

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